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Strategic Analysis and Operational Framework

Strategic Analysis and Operational Framework: Data Quality Analyst II within the Texas Farm Bureau Underwriting Data Office

Historical Foundation and Strategic Context of the Waco Command Center

The operational landscape of the Texas Farm Bureau Casualty Insurance Company as of April 17, 2026, represents a sophisticated convergence of historical agrarian advocacy and the modern, rigorous requirements of property and casualty insurance data management.[1] Based in the Waco home office at 7420 Fish Pond Road, the Underwriting Data Office (UDO) functions as the vital nexus between technical infrastructure and regulatory compliance, particularly in light of the transformative House Bill 2067 mandates.[1] The role of the Data Quality Analyst II is central to this architecture, serving not merely as a technical specialist but as a guardian of the high-fidelity data required to maintain the organization's "Excellent" financial rating and its commitment to over 500,000 member-families.[1]

To evaluate the current data imperatives at Texas Farm Bureau Casualty, it is necessary to examine the provenance of its data assets. Established in 1933 during the economic volatility of the Great Depression, the Texas Farm Bureau emerged as a unified advocate for the state's agricultural producers.[1] The 1938 relocation of headquarters from Dallas to Waco was a strategic maneuver to align the organization with the geographic hub of Central Texas agriculture, ensuring the organization remained physically and culturally proximate to the farmers it served.[1] The subsequent formalization of the Casualty Insurance Company in 1952 established a member-governed structure that prioritizes long-term stability and personal service over external shareholder interests.[1]

As of April 2026, the organization manages a complex ecosystem involving 214 counties and over 850 multi-line agents.[1] The transition from legacy paper-based record-keeping to a centralized digital infrastructure fueled by Guidewire Cloud environments defines the current technical epoch.[1] The accuracy of these records is a prerequisite for justifying rate changes within the Texas "file-and-use" system, where actuarial models must respond to extreme weather loss events that have exceeded $1 billion with increasing frequency over recent decades.[1] Consequently, data quality is viewed as a risk mitigation engine intended to prevent adverse selection and ensure the solvency of the member-focused model.[1]

Organizational Attribute	Metric / Detail (As of April	Source

	2026)	
Year Founded	1933 (Parent); 1952 (Insurance)	[1]
Total Member-Families	538,064	[1]
A.M. Best Rating	A (Excellent); Negative Outlook (as of Sept 2025)	[1]
Agent Force	850+ Multi-line Agents	[1]
County Presence	214 Physical; 205 Self-Governed	[1]
Total Assets (2024)	Approximately $982.5 Million	[1]
Total Revenue (2024)	Approximately $34.2 Million	[8]
Net Assets (2024)	Approximately $968.7 Million	[8]

The strategic context in 2026 is further complicated by severe financial strain within the agricultural sector. A nationwide survey conducted in April 2026 revealed that 70 percent of farmers cannot afford sufficient fertilizer for the 2026 season due to a 30% rise in nitrogen fertilizer prices and a 20-40% increase in combined fuel and fertilizer costs.[9] For an organization like Texas Farm Bureau, which is intrinsically linked to the financial health of producers, these macroeconomic pressures underscore the necessity for precise underwriting and data integrity. The Data Quality Analyst II must ensure that the data reflecting the "Voice of Texas Agriculture" is accurate enough to support the resilient but stressed membership.[4]

The 2026 Regulatory Paradigm: House Bill 2067 and TDI Compliance

The primary challenge facing the Underwriting Data Office in April 2026 is the implementation of House Bill (HB) 2067, passed during the 89th Texas Legislature in 2025.[1] This legislation significantly altered the reporting obligations for property and casualty insurers by replacing the requirement for policyholders to request a written explanation for adverse coverage decisions with an automatic disclosure obligation.[1] Insurers are now legally mandated to proactively provide written reasons for declinations, cancellations, or non-renewals.[1]

The Texas Department of Insurance (TDI) has adopted a phased approach for HB 2067, with Phase 1—effective April 1, 2026—focusing on residential property and private passenger automobile insurance.[1] This regulatory shift transforms the capture of reason codes from a customer service function into a high-rigor data reporting mandate.[1] Under Sections E, F, and G of the revised Texas Statistical Plans, insurers must submit monthly (residential) or quarterly (auto) reports by ZIP code to monitor geographic market trends and underwriting decisions.[1]

Implementation Phase	Coverage Scope	Status / Key Deadlines	Source
Phase 1	Residential Property, Private Auto, Workers' Comp	Effective 4/1/2026; First Report Due 6/15/2026	1
Phase 2	Commercial Lines	Rules proposed in 2026	1
Phase 3	Other P&C lines (Ocean Marine, Umbrella, etc.)	Under investigation as of 4/2026	1

The technical specifications for these reports require a monthly summary of notices sent, including a 60-day indicator (Column 18) and a concatenated alphabetical list of standardized reason codes.[1] Section G requires a count of "actualized actions" that must reconcile with the NAIC Market Conduct Annual Statement (MCAS).[1] A critical operational challenge involves the retroactive data capture requirement: the first report due June 15, 2026, must include non-renewal notices sent before April 1, 2026, if the action effective date falls on or after April 1, 2026.[1]

Standardized Reason Codes for Residential Risk

The Data Quality Analyst II must ensure that reason codes generated by automated underwriting systems are mapped precisely to the TDI's shorthand codes.[1] When multiple reasons apply, the codes must be concatenated in alphabetical order.[1] Discrepancies between the notice sent to the policyholder and the statistical file submitted to TDI are detectable and may lead to regulatory penalties.[1]

Code	Reason Description	Applies To	Source
A	Aggressive or threatening behavior	C, NR, D	1

B	Fraud or material misrepresentation	C, NR, D	[1]
C	Failure to pay premium	C	[1]
D	Underwriting - Physical condition of property	C, NR, D	[1]
E	Underwriting - Geographic/territory restrictions	C, NR, D	[1]
F	Underwriting - Prior claims history	C, NR, D	[1]
G	Underwriting - Occupancy or use of property	C, NR, D	[1]
H	Underwriting - Inability to inspect property	C, NR, D	[11]
I	Underwriting - Risk exceeds capacity	C, NR, D	[11]
J	Underwriting - Insured value or replacement cost	C, NR, D	[11]
K	Underwriting - Roof condition	C, NR, D	[1]
L	Underwriting - All other underwriting reasons	C, NR, D	[1]

M	Policy no longer available in market	NR	[1]
N	Change in insurer's appetite/market withdrawal	NR, D	[1]
O	Insured's request	C	[11]
P	Company/agency relationship change	C, NR	[11]
Q	Insured obtained coverage elsewhere	C	[11]
R	Other	C, NR, D	[11]
X	Assumption Reinsurance (TWIA only)	C	[1]

($C =$ Cancellation, $NR =$ Non-renewal, $D =$ Declination)

The implications of this reporting structure are profound for data integrity. The plan explicitly states that certain claims, such as those caused by natural causes or those prohibited under Texas Insurance Code §544.353, may never be counted as chargeable claims.[11] Furthermore, the amendments to 28 TAC §5.7015, effective September 1, 2026, prohibit insurers from using "short rate" provisions, requiring all unearned premiums to be calculated pro rata.[18] The Analyst serves as the primary auditor of this pipeline, utilizing SQL to reconcile notice tables with statutory submission tables and ensure that consumer disclosures reflect the actual reported codes.[1]

Technological Infrastructure: Guidewire Olos and Informatica IDMC

The Underwriting Data Office operates within a sophisticated technical stack designed to eliminate data silos and manual processes.[1] As of April 17, 2026, the core system is the Olos release of Guidewire PolicyCenter, which reached availability in December 2025.[1] This platform is integrated with Informatica Intelligent Data Management Cloud (IDMC) for data quality orchestration and Microsoft Dynamics 365 for agency relationship management.[1]

Guidewire Olos Release Capabilities

The Olos release introduces critical validation and performance features that represent a significant advancement over previous versions. One primary feature is the "Validation endpoint support for PolicyCenter," allowing for the creation of request-and-response validation flows triggered by application events.[1] This allows the Data Quality Analyst II to implement automated checks that execute before a policy is bound, ensuring that third-party risk scores, such as those from HazardHub, are correctly mapped into the rating algorithm.[1]

Olos also features "Spotter," a conversational search interface (formerly branded as Sage) that utilizes AI to allow underwriters and analysts to query the data model using natural language.[1] This integration of "AI Highlights" provides a strategic advantage by automatically summarizing key drivers behind KPI changes on Liveboards.[1] For instance, the system can identify that a specific ZIP code is seeing an unexpected spike in "Roof Condition" cancellations by analyzing the first five time-series KPIs in a dashboard tab.[1] This automation reduces the turnaround time for data-driven decisions from days to minutes.[1]

Guidewire Olos Feature	Function in Underwriting Data	Technical Impact	Source
Spotter (AI Search)	Natural language queries for data models	Conversational follow-up and accuracy coaching	22
AI Highlights	Automated summary of KPI drivers	Identifies outliers in Liveboard time-series	20
Validation Endpoints	Trigger checks via App Events	Validates connections without payload restructuring	22
Post-Process Endpoints	Solution execution post-results	Parallel processing for third-party integrations	22
APD Conversion	Product definition shifting to cloud	Reduces technical debt and maintenance overhead	6

Informatica Intelligent Data Management Cloud (IDMC)

Informatica IDMC serves as the industrial-strength engine for data cleansing and metadata observability.[1] With the CLAIRE AI engine, IDMC provides automated metadata management and lineage tracking, essential for maintaining the audit compliance required by HB 2067.[1] The transition to IDMC's consumption-based IPU (Informatica Pricing Units) model represents an operational shift for the department, requiring the Analyst to monitor burn rates across different services.[1] This is particularly urgent as standard support for the legacy on-premise PowerCenter ended on March 31, 2026, forcing a mandatory migration to the cloud platform.[26]

Informatica Service	Function in Underwriting Data	Metering Context (IPU)	Source
CDI (Integration)	Real-time data movement to DataHub	Scaled by Secure Agent and CDC usage	1
CDQ (Quality)	Profiling and Rule Execution	Volume-based scaling; higher rate than CDI	1
CLAIRE Engine	AI-powered mapping and recommendations	Consumption-based AI optimization	1
CLAIRE GPT	Generative AI for data tasks	Trial at no cost through Jan 31, 2027	24
MDM SaaS	Master Data Management (Customer 360)	Hybrid model (IPU + per-domain records)	1

The Analyst utilizes IDMC to profile data in real-time as it moves from legacy systems to the Guidewire DataHub, ensuring that Customer 360 initiatives are based on a "golden record" that has been de-duplicated and standardized.[1] This is critical for cross-selling and upselling initiatives, as it allows for a comprehensive view of relationships between customers, policies, and claims.[1]

End-to-End Data Quality Lifecycle: Operational Flow

The Texas Farm Bureau Casualty Data Quality Analyst II framework defines a comprehensive lifecycle for underwriting data, beginning with ingestion and proceeding through profiling, validation, correction, and monitoring.[1] Each step is supported by specific tools and methodologies designed to ensure that data is "fit for purpose" in a highly regulated environment.[1]

Ingestion and Profiling

Data ingestion involves loading raw underwriting data—policy records, exposures, financials—into a staging area or data lake.[1] External reference data, including agent lists, rating manuals, and TDI statistical tables, are also imported.[1] The first operational step is column-level profiling to establish baseline metrics.[1] This involves computing counts of null values, distinct value counts, and outlier detection.[1]

Profiling is not a one-time event but a continuous diagnostic tool.[1] The Analyst compares current metrics against historical baselines to identify "data drift," which may indicate an upstream change in system logic or agent entry behavior.[1] For example, a sudden jump in null rates for "Roof Age" following a system update would be caught during this phase.[1] Effective profiling involves examining the structure, content, and relationships within the data to identify areas needing improvement.[30]

Validation Rule Execution

Based on profiling and business requirements, the Analyst codifies rules that data must satisfy.[1] Validation acts as a preventive quality control framework, establishing systematic gates that only permit appropriate data to enter the system.[32] Common rules in the Underwriting Data Office include:

- **Mandatory Fields:** Ensuring fields like AgentID or CoverageCode are not null.[1]
- **Value Constraints:** Enforcing that premiums are greater than zero and that TDI codes exist within the approved list.[1]
- **Format Standards:** Validating ZIP code patterns and policy number formats using regular expressions.[1]
- **Referential Integrity:** Ensuring that every Policy.AgentID has a corresponding match in the agent master table.[1]
- **Logical Checks:** Verifying relationships, such as policy start dates preceding end dates.[33]

These rules are executed via SQL scripts, Informatica CDQ rules, or Gosu scripts embedded in Guidewire.[1] Any record failing these checks is flagged in an automated issues log for investigation.[1]

Issue Detection and Root Cause Analysis (RCA)

Detection is automated through SQL Server Agent jobs or BI subscriptions that trigger alerts when error rates exceed defined thresholds.[1] Once an issue is flagged, the Analyst performs Root Cause Analysis using techniques like the "5 Whys" or Fishbone (Ishikawa) diagrams.[1] This investigation traces the error back through the pipeline to determine whether the source was a system bug, an ETL transformation error, or a data entry mistake.[1]

Tracing erroneous records requires deep data lineage reviews.[1] The Analyst may use ETL lineage tools or query historical audit logs in Guidewire to identify if the issue arose during a recent policy migration or a specific batch load.[1] The output of this phase is an RCA report identifying the underlying source and a proposed corrective action.[1] Effective RCA focuses on systems rather than individual performance, ensuring that breakdowns in processes are prevented from recurring.[37]

Correction and Remediation

Corrective actions involve either data cleansing or process fixes.[1] Data cleansing may include updating records via SQL or Python to standardize formats or fill missing values using reference lookups.[1] Process fixes involve updating the underlying integration logic or ETL scripts to prevent the issue from recurring.[1]

Remediation Task	Implementation Methodology	Purpose	Source
Standardize Casing	SQL INITCAP(), UPPER()	Consistency in naming/addresses	[1]
Standardize Address	SQL TRIM(), REPLACE()	Improves geocoding and mailing accuracy	[1]
Impute Missing Values	SQL LEFT JOIN on Ref Tables	Enhances completeness for mandatory fields	[1]
Deduplicate Records	SQL GROUP BY, Python drop_duplicates	Ensures data uniqueness (Customer 360)	[1]
Standardize Phone	SQL REPLACE(col, '-', '')	Normalizes communication metadata	[1]

Enhanced Standard Operating Procedures (SOPs) for the Underwriting Data Office

The following detailed SOPs define the operational rhythm and governance standards of the department as of April 17, 2026.[1]

SOP 1: End-to-End Data Quality Lifecycle Management

The primary objective is to ensure all data ingested from source systems is fit for purpose and compliant with TXFB and TDI standards.[1]

1. **Ingestion & Profiling:** Establish an inventory of fields and compute baseline metrics immediately

upon data arrival.[1]

2. **Validation Rule Application:** Use the rules engine to enforce constraints. Rules should be updated dynamically to adapt to new regulations like HB 2067.[1]
3. **Issue Detection & Alerting:** Monitor failures against the 1% Data Quality Acceptance Standard. Rejection of a submission occurs if invalid formats or illogical entries exceed this threshold.[1]
4. **Root Cause Analysis (RCA):** Implement the structured investigation process described in SOP 4.[1]
5. **Cleansing & Remediation:** Execute fixes and document them in the remediation plan, tracking the "First Fix Rate" and "Average Time to Resolution".[1]

SOP 2: Data Profiling and Statistical Assessment

This procedure ensures that the statistical integrity of the underwriting book is maintained through regular diagnostics.[1]

1. **Continuous Profiling:** Establish a schedule (e.g., weekly) for profiling core datasets to identify anomalies or shifts in data patterns.[1]
2. **Metric Comparison:** Compare current metrics (e.g., policy counts by ZIP code) against historical data to identify potential geographic market trends that may interest TDI.[1]
3. **Stakeholder Communication:** Provide data health scorecards to underwriting managers to support data-driven decision-making.[1]

SOP 3: Validation Rule Execution

The objective is to establish proactive gates that prevent "dirty data" from reaching production environments.[1]

1. **Rule Definition:** Collaborate with business analysts to define logical and technical constraints for all new fields (e.g., the "60D" indicator for HB 2067).[16]
2. **Constraint Selection:** Apply "system constrained" rules for mandatory fields (e.g., drop-down lists for reason codes) to minimize manual entry errors.[29]
3. **Threshold Setting:** Define the maximum percentage of records that can fail a specific rule before a job is aborted (e.g., a 5% threshold for non-critical fields).[34]
4. **Logging and Audit:** Maintain detailed logs of every validation execution, including the MIS date (data filter date) and specific iteration ID.[33]

SOP 4: Root Cause Analysis and Remediation

This procedure ensures that systemic issues are identified and eliminated to prevent recurrence.[1]

1. **Problem Statement:** Define the problem using the SMART principle (Specific, Measurable, Action-oriented, Realistic, Time-constrained).[37]
2. **Data Collection:** Gather comprehensive data, including incident reports, system logs, and stakeholder interviews, to establish a timeline of events.[37]
3. **Analysis Phase:** Use the "5 Whys" method or Ishikawa diagrams to hypothesize and identify the

fundamental breakdown in the process.[1]

4. **Corrective Action Implementation:** Develop an action plan that outlines steps, assigns responsibilities, and sets deadlines.[35]
5. **Documentation:** Create a comprehensive "post-mortem" report detailing the RCA process, root causes identified, and effectiveness of the solution.[37]

SOP 5: Data Issue Management and Severity Escalation

This procedure defines the response protocol based on the severity of the data quality event.[1]

1. **Severity Classification:**
   - **Critical:** Issues halting underwriting or breaching TDI SLAs (e.g., HB 2067 reporting failure). Requires 1-4 hour investigation.[1]
   - **High:** Errors affecting pricing models or risk assessment (e.g., HazardHub failure). Requires 24-hour resolution.[1]
   - **Medium/Low:** Reporting inaccuracies or cosmetic hygiene.[1]
2. **Stakeholder Notification:** Alert the Underwriting Manager and PMO for all Critical and High incidents.[1]
3. **Resolution Tracking:** Document root cause, fix steps, and preventive measures before closing the ticket.[1]

SOP 6: Data Quality KPI Monitoring and Reporting

This procedure focuses on the systematic evaluation of performance metrics to maintain long-term data health.[1]

1. **KPI Establishment:** Define targets and thresholds for accuracy, completeness, consistency, timeliness, validity, and integrity.[1]
2. **Automated Monitoring:** Utilize BI platforms (like D2D) to host KPIs and set event-driven notifications for threshold breaches.[36]
3. **Performance Auditing:** Conduct regular reviews of agency-specific KPIs to ensure data meets "fitness for purpose" requirements.[1]
4. **Reporting Cadence:** Generate monthly reports summarizing data quality trends and the effectiveness of remediation plans.[35]

SOP 7: Data Governance Execution and Data Dictionary Maintenance

Embedding DAMA-DMBOK principles into daily operations preserves metadata integrity.[1]

1. **Dictionary Updates:** When new fields are added to Guidewire or Dynamics 365, update the central Data Dictionary with the field name, definition, format, and ownership.[1]
2. **Naming Standards:** Enforce consistent naming conventions (e.g., YYYYMMDD for dates) across all datasets.[1]
3. **Data Inventory:** Maintain an accurate Record of Processing Activities (RoPA) to track data

categories, retention periods, and technical safeguards, as required for GDPR/DORA compliance.[39]

SOP 8: SDLC Support and Release Readiness

The objective is to ensure data integrity is maintained through system updates and cloud releases.[1]

1. **Release Assessment:** Review release notes for platforms like Guidewire Olos to identify new data model changes or API endpoints.[20]
2. **Test Case Development:** Design validation tests to verify that new features (e.g., Spotter conversational search) are accurately reflecting the underlying data.[1]
3. **Regression Testing:** Ensure existing validation rules are not compromised by new system code or configuration overrides.[1]

Advanced SQL Methodology for Data Quality Analysis

The mastery of SQL is the foundational skill for a Data Quality Analyst II, extending beyond retrieval to the application of set theory and complex transformations.[1]

The Application of Set Theory

In the context of property and casualty insurance, policies, periods, and locations maintain complex one-to-many relationships.[1] Set theory is functionally applied during data reconciliation tasks, particularly after nightly batch loads. The Analyst must ensure that the set of policies in the staging environment (S) is identical to the set of policies processed in the production environment (P), identifying symmetric differences that indicate pipeline failures or record drops.[1]

$$\Delta = (S \setminus P) \cup (P \setminus S)$$

Aggregate Profiling for Data Health

Aggregate profiling serves as the primary diagnostic tool for monitoring ongoing data integrity.[1] By utilizing complex CASE logic within aggregate functions, the Analyst can quantify the health of millions of records in a single execution.[1] This is critical for computing completeness metrics for mandated fields like "Roof Coverage Type" or "VIN."

SQL Logic for Completeness Indexing:

SQL

```
CompletenessIndex =
(SUM(CASE WHEN VIN IS NOT NULL AND LEN(TRIM(VIN)) = 17 THEN 1 ELSE 0 END) * 100.0) /
```

COUNT(*)

This logic allows the Underwriting Data Office to report to stakeholders exactly what percentage of the book of business meets the necessary underwriting standards.[1]

Exhaustive SQL Library for Underwriting Data Quality

The following SQL queries represent the technical standard for the role at Texas Farm Bureau as of April 17, 2026.[1]

1. Statutory Compliance Audit: HB 2067 Reason Code Reconciliation

This query identifies policies where the cancellation reason stored in PolicyCenter does not align with the mandatory TDI codeset.[1]

SQL

```sql
SELECT
    p.PolicyNumber,
    p.CancellationReasonCode AS Internal_Reason,
    t.TDICode AS Statutory_Reason,
    CASE
        WHEN t.TDICode IS NULL THEN 'Missing Mapping'
        WHEN p.CancellationReasonCode!= t.TDICode THEN 'Mapping Discrepancy'
        ELSE 'Valid'
    END AS Compliance_Status
FROM pc_policyperiod p
LEFT JOIN TDI_Reason_Mapping t ON p.CancellationReasonCode = t.InternalCode
WHERE p.Status = 'Canceled'
  AND p.CancellationDate >= '2026-04-01'
  AND p.IsMostRecentModel = 1;
```

2. Data Integrity: Detecting "Orphan" Risks

This query detects risk locations or coverages that lack a parent PolicyPeriod, a common issue following complex endorsements or system migrations.[1]

SQL

```
SELECT
  l.PublicID AS Location_ID,
  l.LocationNumber,
  p.PolicyNumber
FROM pc_policylocation l
LEFT JOIN pc_policyperiod p ON l.BranchID = p.ID
WHERE p.ID IS NULL
 OR p.Status IN ('Draft', 'Withdrawn');
```

3. Data Cleansing: Normalizing Vehicle Identification Numbers (VIN)

Standardizing unstructured text is a core responsibility. This script identifies invalid VIN lengths and removes non-alphanumeric characters.[1]

SQL

```
SELECT
  PolicyNumber,
  VIN,
  LEN(VIN) AS VIN_Length
FROM pc_vehicle
WHERE LEN(VIN)!= 17
  OR VIN LIKE '%[^a-zA-Z0-9]%';
```

4. Geocoding Accuracy Audit for Coastal Risk

Ensures that risks in high-exposure coastal counties have valid geographic coordinates for catastrophe modeling.[1]

SQL

```
SELECT
  p.PolicyNumber,
  l.AddressLine1,
  l.Latitude,
  l.Longitude
FROM pc_policylocation l
JOIN pc_policyperiod p ON l.BranchID = p.ID
```

```sql
WHERE p.IsMostRecentModel = 1
  AND l.State = 'TX'
  AND (l.Latitude IS NULL OR l.Longitude IS NULL)
  AND l.County IN ('Harris', 'Galveston', 'Nueces', 'Cameron');
```

5. Aggregate Profiling: Monitoring Daily Ingest Volume Spikes

Identifies anomalies in ingestion volume that may indicate duplicate batch runs or feed failures.[1]

SQL

```sql
SELECT
    CAST(CreateTime AS DATE) AS Ingest_Date,
    COUNT(*) AS Row_Count,
    AVG(COUNT(*)) OVER (ORDER BY CAST(CreateTime AS DATE) ROWS BETWEEN 7 PRECEDING
AND 1 PRECEDING) AS Seven_Day_Avg
FROM pc_policyperiod
GROUP BY CAST(CreateTime AS DATE)
HAVING COUNT(*) > 1.25 * (AVG(COUNT(*)) OVER (ORDER BY CAST(CreateTime AS DATE) ROWS
BETWEEN 7 PRECEDING AND 1 PRECEDING));
```

6. Cross-System Reconciliation: PolicyCenter vs. BillingCenter

Ensures consistent data flow between the policy and billing systems, essential for financial accuracy.[1]

SQL

```sql
SELECT
    p.PolicyNumber,
    p.TotalPremium AS PC_Premium,
    b.TotalPremium AS BC_Premium
FROM pc_policyperiod p
JOIN bc_policy b ON p.PolicyNumber = b.PolicyNumber
WHERE p.IsMostRecentModel = 1
  AND p.TotalPremium!= b.TotalPremium;
```

Operational Rhythm and Failure Response Playbooks

To maintain a consistent standard of data health, the Underwriting Data Office follows a defined operational cadence as of April 2026.[1]

Daily/Weekly Cadence

The Analyst starts every morning by reviewing the Data Quality Dashboards in Power BI and addressing alerts triggered by SQL Server Agent jobs.[1] Daily tasks include investigating critical exceptions and updating the team on resolution status during the morning stand-up.[1] Weekly, the Analyst conducts in-depth reviews with Underwriting and IT to discuss trending issues, update KPIs, and share summary reports with business owners.[1] Performance self-reviews are typically conducted on a weekly basis, identifying training needs and entry trends.[43]

Failure Playbooks

Predefined protocols are established for common failure scenarios to minimize operational impact.[1]

- **ETL Job Failures:** If an overnight load fails, the Analyst notifies data engineers and attempts an automated restart.[1] Downstream reporting is temporarily halted to prevent the use of stale or incomplete data.[1]
- **Sudden Error Spikes:** A jump in error rate (e.g., exceeding the 1% acceptance standard) triggers an immediate SOP 4 investigation to isolate contributing factors and contain the problem.[1]
- **Regulatory Non-Compliance:** If reason codes are found to be missing or misaligned for HB 2067 notices, the issue is escalated to the "Critical" severity level, and a mandatory remediation plan is initiated with TDI notification if required.[1]

Data Quality Dimensions and Performance Metrics

The success of the Data Quality Analyst II is measured through a set of operationalized KPIs based on the six dimensions of data quality: Accuracy, Completeness, Consistency, Timeliness, Validity, and Integrity.[1]

Dimension	Metric	Departmental Target	Source
Accuracy	% of records failing a quality rule	< 1.0% Error Rate	[1]
Completeness	Fill rate for mandatory TDI fields	100% (No Nulls)	[1]
Consistency	Mismatch count between PC and Billing	0 Discrepancies	[1]

Timeliness	Average lag in nightly data loads	< 4 Hours	[1]
Validity	% of values conforming to TDI code lists	100%	[1]
Integrity	Count of "Orphan" records in staging	0 Orphans	[1]

These KPIs are tied to performance management.[1] Each data domain has an assigned Data Owner who is responsible for the health of their relevant metrics.[1] Dashboards provide real-time feedback, encouraging teams to remediate issues proactively rather than waiting for statutory audits.[1] High-quality data reduces the number of errors encountered, improves customer relations, and keeps the company prepared for the increased oversight of TDI and AM Best.[3]

Advanced Technical Applications: Agentic Data Management

The strategic direction for the Texas Farm Bureau Underwriting Data Office as of April 2026 involves the adoption of "Agentic" data management, leveraging the Guidewire Agentic Framework and Informatica's AI-powered services.[1]

Autonomous Remediation and Risk Mitigation

Agentic systems, such as those integrated into Informatica IDMC and Guidewire Olos, can autonomously prioritize the most impactful data fixes based on real context.[1] For example, if an AI agent detects a high-impact error in "Roof Coverage Type" for coastal risks, it can suggest a remediation path, verify the fix across related systems, and update the audit trail—all with minimal human intervention.[1] The Guidewire Rules Service enables centralized rules administration through a visual Rules Designer, allowing the UDO to respond to new requirement changes without code deployment delays.[21] This shift fundamentally optimizes cost structures and recaptures time previously lost to manual remediation.[1]

Human-in-the-Loop Governance

The role of the Data Quality Analyst II evolves from manual monitoring to presiding over these autonomous systems.[1] Governance frameworks must include clear escalation protocols: defining which decisions can be fully automated (e.g., standardizing address casing) and which require human approval (e.g., modifying reason code concatenation logic for HB 2067).[1] "Garbage In, Garbage Out" (GIGO) remains a critical principle; the Analyst must ensure that the foundational data architecture is robust enough for these agents to operate accurately.[1] This requires personnel to be highly trained in root cause investigation techniques and advanced validation methodologies to oversee the "Agentic" lifecycle.[6]

The convergence of high-rigor regulatory mandates, economic pressures on the membership, and advanced AI-driven technology necessitates a Data Quality Analyst II who is not just a technician but a data architect of integrity.[1] By maintaining these exhaustive standards and procedures, the Underwriting Data Office ensures that Texas Farm Bureau Casualty Insurance Company remains the "Voice of Texas Agriculture" in an increasingly data-dependent world.[1]

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